

ARUNKUMAR K R

✉ arun8778jul@gmail.com | ☎ +91 87787 65734 | 📁 Portfolio | in LinkedIn
🐙 GitHub | 💻 Leetcode | 🏠 Hackerrank

SUMMARY

Computer Science Engineering undergraduate with internship experience in Full Stack Development and Machine Learning Engineering. Skilled in Python, Java, React.js, Node.js, FastAPI, PostgreSQL, Docker, REST APIs, and LLM-based systems. Built scalable web applications, AI pipelines, and backend systems with measurable performance improvements. Published researcher in Computer Vision and RHCSA-certified.

EDUCATION

KPR Institute of Engineering and Technology, Coimbatore	2023 – 2027
B.E. Computer Science and Engineering	CGPA: 8.77 / 10
Kongu Vellalar Matriculation Higher Secondary School, Perundurai	2022 – 2023
Higher Secondary Education (Class XII)	94.6%

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript (ES6+), C, SQL
- **Frameworks & Libraries:** React.js, Node.js, Express.js, FastAPI, Flask, Streamlit, LangChain, TensorFlow, PyTorch, Scikit-Learn, Pandas, Numpy
- **Tools & Platforms:** Git, GitHub, Docker, REST APIs, Linux, RHCSA, PowerBI
- **Core Concepts & Databases:** DSA, OOP, DBMS, Operating Systems, PostgreSQL, MongoDB, ChromaDB, Redis

EXPERIENCE

Reccsar Pvt Ltd	Jan 2025 – Feb 2025
<i>Full Stack Developer Intern</i>	
• Developed 5+ production-ready full-stack features using React.js, Node.js, and Express.js, improving usability and supporting live product workflows for 100+ users. • Integrated 10+ REST APIs with PostgreSQL-backed services to build dynamic UI components, reducing manual data entry effort by approximately 30%.	

PROJECTS

ProManage – Collaborative Project Management API	Live Demo GitHub
<i>Node.js, TypeScript, PostgreSQL, Redis, JWT</i>	
• Architected a scalable REST API with Role-Based Access Control (RBAC) and JWT authentication, enforcing granular permission tiers across user roles (admin, manager, member). • Reduced API response time by 45% using Redis caching and Node.js Streams, enabling low-latency performance under 500+ concurrent simulated requests.	
Prism AI – Content Generation Suite	Live Demo GitHub
<i>Python, FastAPI, Groq LLM, Hugging Face FLUX.1</i>	
• Built a multi-modal AI platform using FastAPI, Groq LLM, and Hugging Face FLUX to generate SEO blogs, scripts, and images, supporting 3 content types through asynchronous APIs. • Served a glassmorphism single-page frontend through a unified FastAPI backend, reducing infrastructure complexity while supporting multiple concurrent AI generation tasks.	
RouteX – Intelligent PDF Routing System	Live Demo GitHub
<i>Python, PyMuPDF, Tesseract OCR, NLP</i>	
• Developed a PDF routing pipeline using PyMuPDF, Tesseract OCR, and NLP that classified scanned and digital documents with 95% accuracy. • Applied NLP-based summarization and TF-IDF keyword extraction for automatic document tagging, reducing manual categorization effort significantly.	

RESEARCH & ACHIEVEMENTS

- **Research Publication:** "Real-Time Surface Defect Detection in Metals using YOLOv8" – presented and published at the **3rd ICEAMST International Conference**; focused on industrial CV applications with deep learning.
- **RedHat Certified System Administrator (RHCSA)** – Industry-recognized Linux system certification.
- **Competitive Programming:** Solved **500+ DSA problems** on LeetCode and HackerRank, with verified certifications in Problem Solving (Intermediate), Python, Java, and SQL.
- **Leadership:** Executive Member at Dept. Association – coordinated technical events and workshops for 100+ students.